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An extreme value analysis of the tail relationships between returns and volumes for high frequency cryptocurrencies

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ABSTRACT

This paper investigates both the extreme dependence and correlation between high frequency cryptocurrency (Bitcoin and Ethereum, versus the Euro and US Dollar) returns and transaction volumes, at the extreme tails associated with booms and busts in the cryptocurrency markets. We apply an extreme value theory (EVT) approach, and highlight how these results assist traders and practitioners who rely on such technical indicators in their trading strategies – especially in times of extreme market turbulence or irrational market exuberance. Our findings contradict the belief in Wall Street that volume can significantly influence price levels and from an economic perspective our model reveals weak positive correlation between return and volume at the tails, which suggests that a misinterpretation among market participants can cause cryptocurrency markets to be relatively illiquid, thus leading to extreme price movements. Relating our statistical findings to economic models, we find that our empirical results are consistent with the explanation of market crashes based on trade misinterpretation.

1. Introduction

The dynamic volume–return relationship conveys valuable information about future price movements for market participants, this makes it an important factor when studying the market microstructure. This information can be used in the analysis of market efficiency, price prediction, and the development of trading strategies. In general, the literature on the stylised facts of volume and return suggests that there is a positive correlation between returns and absolute return volume in market indices (Jain and Joh, 1988), commodities, equities (Ying, 1966; Llorente et al., 2002), interest rates, currency futures (Puri and Philippatos, 2008), and real estate (Tsai, 2014). More specifically, Llorente et al. (2002) suggest that speculative trading has an asymmetric effect on different types of assets, which leads to variation in their volume–return dynamics. However, this relationship is more generally influenced by factors driving trading. Brailsford (1996) also supports this asymmetric effect and suggests that this is due to the differential costs of opening long and short trading positions respectively. With the recent market turbulence in financial markets, the importance of examining the tail relation between return and volume at the extremes tails is essential to better understand the behaviour of market participants during extreme events. Longin and Pagliardi (2016) apply extreme value theory to investigate the relationship between transaction volume and returns in the tails of the distribution associated with booms and crashes in the US stock market. Results show that extreme correlation between return and volume decreases as we consider larger events in both tails of the distributions. The return-volume

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relationship for seven large international equity markets was investigated by Marsh and Wagner (2004) in times of market stress. The results reveal that trading volume is vital in assisting the understanding of expected equity returns, return variability, and large price movements. Both Gennotte and Leland (1990) and Balduzzi et al. (1996) showed that misinterpretation among market participants can cause financial markets to be relatively illiquid, leading to extreme price movements. The results showed that the extreme correlation between returns and volume decreases at the extreme tails.

Cryptocurrencies are digital currencies that utilise blockchain technology along with cryptography to provide secure, anonymous transactions for users. Initially developed predominantly for speed and economic efficiency when transferring money globally, they have evolved, becoming a much more versatile asset. The cryptocurrency market currently comprises over 11,000 (CoinMarketCap, 2020) different currencies, with Bitcoin and Ethereum being the two largest, whose main purposes include faster and more efficient monetary payments, decentralised finance (DeFi), decentralised applications (DApps), and smart contracts (Zhang et al., 2019b). As noted by Shi and Shi (2021) and Chu et al. (2021), the existing literature has covered Bitcoin and other cryptocurrencies extensively from an economic and financial perspective. The majority investigate classical financial economics, for example, the general price discovery and predictability (Godfrey, 2017; Brauneis and Mestel, 2018); the informational efficiency (Urquhart, 2016; Bariviera, 2017; Nadarajah and Chu, 2017); liquidity (Balcilar et al., 2017; Loi, 2017; Zhang et al., 2020); volatility (Brandvold et al., 2015; Dyhrberg, 2016; Katsiampa, 2017; Urquhart, 2017); Bitcoin system (Gandal et al., 2018); perspective of a currency or an asset (Yermack, 2015; Sapuric and Kokkinaki, 2014; Polasik et al., 2015) and market regulations (Hendrickson and Luther, 2017; Pieters and Vivanco, 2017).

Despite these numerous studies on cryptocurrencies, the analysis of the volume–return relation in cryptocurrencies is scarce and there appears to be little research. The few studies that do exist include Balcilar et al. (2017) who employed non-parametric causality-in-quantiles tests to analyse the causal relationship between trading volume and Bitcoin returns and volatility. Showing that volume can predict returns, except during bull and bear periods in the Bitcoin market. Another, Zhang et al. (2018), who investigated the return-volume relationship of the Bitcoin market based on multifractal detrended cross-correlation analysis found that Bitcoin exhibits a non-linear dependent relationship in return-volume with cross-correlations of return-volume showing anti-persistent behaviour. Finally, Naeem et al. (2020) implemented copulas to measure the dependence between extreme returns and volume among different cryptocurrencies and discovered stronger dependence for the lower tail than the upper tail. However, no study has investigated this relationship at high frequency in cryptocurrencies. Hence, our main motivation is to contribute to the literature by using extreme value theory – the extreme volume–return dependence in cryptocurrencies at high frequency. The importance of high-frequency cryptocurrency trading was noted by Chu et al. (2019) "it was not too long ago when individuals bought and sold cryptocurrencies monthly and weekly – making small profits here and there. However, more recently, many online exchanges have started offering tools and services that allow users to trade cryptocurrencies at a very high speed over the course of minutes and seconds". Hence, the study of high-frequency cryptocurrency data (e.g. hourly prices and returns) is becoming increasingly important.

For many traders and practitioners, the lack of available fundamental valuation on cryptocurrencies creates a barrier preventing them from applying traditional valuation techniques to quantify the true intrinsic value of cryptocurrencies. This has led to many relying on technical analysis (Hudson and Urquhart, 2019) or utility valuation as an alternative method to analysing cryptocurrencies prices. Understanding the dynamics of volume and returns is essential from an economic and financial perspective in acknowledging how market information is conveyed and incorporated in asset prices. In addition, in times of extreme market turbulence or market exuberance, it is vital to examine the volume–return relationship to enhance the understanding of traders and practitioners who rely on these technical indicators in their trading strategies. This is especially when analysing periods of market booms and crashes (extreme events). The fact that investors and traders rely heavily on models based on trading rules (and the relation between return and volume), in addition to the availability of high-frequency cryptocurrency data and shift towards high-frequency trading, further stresses the need for a better understanding of the volume–return relationship in cryptocurrencies.

There is already an extensive literature analysing cryptocurrencies at high frequencies from many different angles. In particular, many researchers have investigated the market efficiency and liquidity of high frequency cryptocurrency markets (Chu et al., 2019 and Zhang et al., 2019b, 2020). Furthermore, the understanding of the microstructure of cryptocurrencies at high frequencies was examined by Stavroyiannis et al. (2019) who examined Bitcoin's multifractal properties using high frequency data; Scaillet et al. (2020) who studied the impact of microstructure and its effect on high frequency jumps in the Bitcoin market; Zhang et al. (2019c) who implemented extreme value theory to investigate the tail risk of high frequency (hourly) returns of cryptocurrencies; Eross et al. (2019) examined the intraday dynamics of the stylised facts and the interactions among returns, volume, bid-ask spread, and volatility of Bitcoin. More recently, the study on the importance of Bitcoin's Futures prices at high frequency was investigated by Baur and Dimpfl (2019) and Fassas et al. (2020) in providing useful information for the price discovery of Bitcoin. In terms of high frequency trading strategies, Chu et al. (2020) analysed and showed potential profitability of using the momentum strategy with cryptocurrencies in a high frequency setting. Hence, our main motivation is to contribute to the literature by examining using extreme value theory, the extreme volume–return dependence in cryptocurrencies at high frequency.

The main contributions of this paper are to (i) address the research gap on extreme dependence between volume and returns at high frequency in cryptocurrencies, and investigate the extreme correlation between the two variables; (ii) apply extreme value theory to investigate the tail relation between the hourly returns and transaction volume of Bitcoin and Ethereum; (iii) assess how these results assist in the trading strategies of traders and practitioners during extreme events.

The contents of this paper are organised as follows. In Section 2, we present some theoretical results of extreme value theory in univariate and multivariate cases. In Section 3, we describe the high-frequency data used in the analysis and provide some summary statistics of the data. Section 4 presents the main results of the extreme correlation between returns and volumes at the extremes and the optimal threshold, while Section 5 implements robustness checks on our models. An economic discussion is provided in Section 6 and finally Section 7 concludes.

2. Modelling approach

In this section, we introduce two extreme value theory tools needed in Section 4. The first approach is to fit a generalised Pareto distribution model to analyse the tails of each marginal distribution. The second is a model to quantify the dependence structure of the extreme values of the two variables using a Gumbel copula.

2.1. Univariate case: modelling the tails of the distributions

Let X denote the log returns of the US dollar exchange rates of a cryptocurrency and Y denote the de-trended traded volume in US dollars. We implement the method of peaks over threshold (POT) to analyse extreme returns and volumes as an exceedance with reference to a threshold denoted by θ. All volumes and returns that lie above (below) θ are categorised as positive (negative) extremes. Let θ^{ret} and θ^{vol} denote the threshold of returns and volumes, respectively. All returns greater than θ^{ret} correspond to a tail probability p of the distribution of returns. The tail probability is derived using the empirical probability of returns greater than the threshold of interest, computed as N/n, where n denotes the total number of observations in the dataset and N is the random number of observations above the selected threshold. It follows that the same tail probability p is used to compute the threshold for volumes θ^{vol}.

The cumulative distribution associated with a random variable X over a threshold θ is denoted by $F_x^{\theta}(x)$.

Pickands (1975) showed that if certain regularity conditions are satisfied and θ is sufficiently large, then $F_x^{\theta}(x)$ can be asymptotically distributed by the generalised Pareto distribution

$$Pr(X > x + \theta | X > \theta) \approx \left[1 + \xi \frac{x}{\sigma}\right]^{-1/\xi}, \tag{1}$$

where $\sigma > 0$ is the scale parameter, $-\infty < \xi < \infty$ is the shape parameter, and $1 + \xi x/\sigma > 0$. Rearranging (1), we can express the distribution function of X as

$$F(x) = Pr(X < x) \approx 1 - p\left[1 + \xi \frac{x - \theta}{\sigma}\right]^{-1/\xi} \tag{2}$$

for $\theta \leq x < \infty$ if $\xi \geq 0$ and $\theta \leq x \leq \theta + \sigma/\xi$ if $\xi < 0$, where $p = \Pr(X > \theta)$.

2.1.1. Determining the optimal threshold

In general, we can choose an arbitrary number above or below the mean to be used as the threshold (θ). However, such an approach can lead to the problem of inefficiency and bias, for instance choosing a large θ will result in few observations of return exceedances, and this results in inefficient parameter estimates with large standard errors. On the contrary, selecting a small θ leads to many observations of return exceedances, but induces biased parameter estimates, as observations not belonging to the tails are included in the estimation process. Hence, the choice of the optimal threshold is a critical issue. To compute the optimal threshold we follow the method as presented in Longin and Solnik (2001) to determine the optimal number of return exceedances. This method assumes that returns can be described by a particular (but simple) model – the Student's t-distribution with some k degrees of freedom. This distribution is chosen for convenience, as the tail index parameter ξ in the generalised Pareto distribution (Eq. (1)), is related to the degrees of freedom parameter k by the relation $\xi = 1/k$. Using this relationship, it follows that the mean squared error (MSE) can be computed in order to find the optimal number of return exceedances and thus the optimal return threshold to be used. The process for finding the optimal number of exceedances and optimal positive threshold is described as follows:

1. Let $k = 1$, generate $S = 500$ simulations of N (total sample size) returns from the Student's t-distribution with k degrees of freedom.
2. For the chosen value of k, consider a range of exceedance values $n = 0.0005 \cdot N, 0.001 \cdot N, \ldots, 0.2 \cdot N$. For each n, compute S estimates $\widehat{\xi}(n, k)$ of the tail index using the largest n returns.
3. For each value of n, compute the MSE of the S tail index estimates as

$$\mathrm{MSE}\left(\widehat{\xi}_{s=1,\ldots,S}\right) = \left(\overline{\widehat{\xi}} - \xi\right)^2 + \frac{1}{S}\sum_{s=1}^{S=500}\left(\widehat{\xi}_s - \xi\right)^2,$$

   and obtain the optimal number of exceedances $n^*(k)$ as the value of n that minimises the MSE for the chosen value of k.
4. Repeat steps 1 to 3, for values of $k = 2, \ldots, 10$, until the optimal number of exceedances $n^*(k)$ have been obtained for $k = 1, \ldots, 10$.
5. For each of the optimal number of exceedances $n^*(k)$, compute an estimate of the tail index parameter $\widehat{\xi}(n^*(k))$ using the $n^*(k)$ largest returns.
6. The true optimal number of exceedances for the distribution of the true returns n^* is given by the $n^*(k)$ value for which $\widehat{\xi}(n^*(k))$ is closest to $\xi = 1/k$.
7. It follows that the optimal threshold value is given as the $1 - (n^*/N)$ quantile of the returns, where the optimal θ^{ret} expresses this value as a percentage above or below the mean of value of the true returns.

Note that the process for finding the optimal negative threshold is as above, but using the negative transformation of the returns.

2.2. Bivariate case: modelling the dependence structure

In order to quantify the dependence structure between extreme values of X and Y, we model their joint cumulative distribution function by a Gumbel (1960) copula function:

$$\begin{aligned} F_{X,Y}(x,y) &= C_\alpha(F_{X_1}(x_1), F_{X_2}(x_2)) \\ &= exp\left\{-\left[\left\{-log\left[1-Pr(X>\theta_1)\left(1+\xi_X\frac{x-\theta_1}{\sigma_X}\right)^{-1/\xi_X}\right]\right\}^{1/\alpha}\right.\right. \\ &\quad \left.\left.+\left\{-log\left[1-Pr(Y>\theta_2)\left(1+\xi_Y\frac{y-\theta_2}{\sigma_Y}\right)^{-1/\xi_Y}\right]\right\}^{1/\alpha}\right]^\alpha\right\} \end{aligned} \tag{3}$$

for $X>\theta_1$ and $Y>\theta_2$, where θ_1 and θ_2 are large thresholds, $\alpha\in(0,1)$ is a dependence parameter, (ξ_X,σ_X) are parameters describing the marginal distribution of X and (ξ_Y,σ_Y) are parameters describing the marginal distribution of Y.

The Gumbel copula model was fitted to the data by the method of maximum likelihood with censored data as in Longin and Solnik (2001), using R (R Development Core Team, 2020) and the package POT. The correlation coefficient ρ between extreme values of X and Y is related to the coefficient α by (Tiago de Oliveira, 1973):

$$\rho = 1-\alpha^2, \tag{4}$$

where ρ takes any value between -1 (strong negative correlation) and $+1$ (strong positive correlation). The special cases of $\alpha=0$ and $\alpha=1$ correspond with total dependence and asymptotic independence, respectively.

3. Data

The data we analyse consists of the historical high frequency hourly returns and trading volumes (the volume of units of cryptocurrency exchanged and the Dollar-volume or Euro-volume of cryptocurrency exchanged), of Bitcoin and Ethereum versus the US Dollar (USD) and Euro (EUR), respectively, listed on the Kraken cryptocurrency exchange, from 11am on 1st July 2017 to 11 AM on 20th April 2020. In order to obtain stationarity of returns and volumes, we applied the data adjustment procedure as in Gallant et al. (1992). The augmented Dickey–Fuller (ADF) test was performed on all of the transformed returns and volumes series, which confirmed stationarity at the 1% level of significance. The cryptocurrencies were chosen based on their ranking in terms of market capitalisation, volume traded, and ease and availability of purchase – of which, Bitcoin (BTC) and Ethereum (ETH) are considered to be the top two. The combined market capitalisation at the start date of our sample data was approximately 80% (CryptoCompare., 2020), thus we believe that the data obtained provides a satisfactory representation of the main market players. In comparison to other financial instruments, cryptocurrencies exhibit greater intra-day volatility (a 3–5% movement is common), and there is evidence that a great proportion of high frequency trades are made through algorithmic trading bots, highlighting the importance of analysing hourly frequency data of cryptocurrencies. For further details regarding the two cryptocurrencies, we refer the readers to Chan et al. (2017) and Zhang et al. (2019a,b).

Summary statistics of the hourly log returns of the cryptocurrencies versus the Euro and US Dollar are given in Table 1. The statistics computed include the number of observations (N), minimum, maximum, mean, median, skewness, kurtosis, standard deviation (SD), variance, range, and interquartile range. From Table 1, we find that the ETH/USD returns give the lowest minimum and

Table 1
Summary statistics of the log returns of the hourly prices of BTC/USD, BTC/EUR, ETH/USD, and ETH/EUR for the period of 11:00 on 01/07/2017 to 11:00 on 20/04/2020, inclusive.

Statistics	BTC/USD	BTC/EUR	ETH/USD	ETH/EUR
Observations	24,434	24,434	24,434	24,434
Minimum	−0.191	−0.173	−0.229	−0.204
Q1	−0.003	−0.003	−0.004	−0.004
Median	0.000	0.000	0.000	0.000
Mean	0.000	0.000	0.000	0.000
Q3	0.003	0.003	0.004	0.004
Maximum	0.174	0.164	0.167	0.166
Skewness	−0.128	−0.005	−0.011	0.112
Kurtosis	30.881	28.568	25.119	24.789
SD	0.010	0.010	0.012	0.012
Variance	0.000	0.000	0.000	0.000
Range	0.366	0.337	0.396	0.370
IQR	0.006	0.006	0.009	0.008

BTC/USD the largest maximum. The median, mean and variance of all pairs of cryptocurrencies are zero. For the returns of all four price series, the kurtosis values are significantly larger than three – indicating that the returns are more peaked and heavier tailed than the normal distribution. However, the returns of three price series (BTC/USD, BTC/EUR and ETH/USD) are negatively skewed, whereas the ETH/EUR returns exhibit positive skewness. With respect to the variability, all returns exhibit similar standard deviations (approximately 1% volatility), whilst the returns of the BTC currency pairs generate the smallest of all these values.

Similarly, summary statistics for the de-trended traded volume expressed in terms of the Euro and US Dollar are given in Table 2. It can be seen that volume exhibits greater volatility than the returns of the cryptocurrencies. Also, for all six traded volume series, the kurtosis values are smaller than those of the returns, which reveals that the distributions of all traded volumes have a lighter tail than the returns. Furthermore, we observe that all traded volume series are positively skewed, which is not the case for the returns.

4. Empirical results

Tables 3–6 present the parameter estimates of return and volume exceedances for Bitcoin (USD and Euro) and Ethereum (USD and Euro) respectively. For each table, Part A gives the results for negative return exceedances (returns below the threshold θ^{ret}) and positive volume exceedances (volumes above the threshold θ^{vol}), and Part B gives the results for positive return exceedances and positive volume exceedances. The threshold levels θ^{ret} that correspond to return exceedances were selected as $\pm$0%, $\pm$1%, $\pm$2%, $\pm$3%, $\pm$4% and $\pm$5% away from the empirical mean of the returns of each currency pair. As our analysis uses log returns, a threshold exceedance of $\pm$1% corresponds to $\pm$0.01 from the empirical mean. The optimal thresholds are also considered in the last row of each table, as computed using the procedures discussed in Section 2.1.1. We observe that the optimal thresholds for both the left and right tails of the return and volume distributions are not identical. In addition, in each table we address the following parameter estimates: the tail probability p, the dispersion parameter σ and the tail index ξ for both returns and volumes, the dependence parameter α of the Gumbel copula and the extreme correlation ρ. Standard errors (where applicable) are given below the estimates in parentheses and were computed using the delta method.

The estimation results for the univariate and bivariate models for BTC/USD return and volume exceedances are shown in Table 3. For Parts A and B in Table 3, we notice that as we move to the extreme left and right tails of the distributions (through using larger fixed thresholds), the dependence parameter between the extreme returns and volumes increases, whilst the extreme correlation decreases. For the negative return exceedances, the optimal threshold is −1.97% for returns and 1.718 for volume, which corresponds to a tail probability of 2.7%. This probability is computed as the number of negative tail observations corresponding with the optimal threshold values (660) divided by the total number of return observations (24,434), and the extreme correlation is equal to 0.126. When considering the positive return exceedances, the optimal threshold is +1.85% for returns and 1.710 for volume, which corresponds to a tail probability of 2.8%, with the extreme correlation being equal to 0.132.

A similar conclusion can be drawn for the estimation results for the univariate and bivariate models for BTC/EUR return and volume exceedances observed in Table 4. However, the optimal threshold values for both the negative/positive return and volume exceedances are smaller than those for the case of BTC/USD. For a visual comparison of the structure of the extreme correlation in Tables 3 and 4, we refer to Figs. 1 and 2, respectively. Figs. 1 and 2 clearly show the structure of the extreme correlation for the fixed thresholds used to define extreme returns ranging from −5% to 0% for negative returns and 0% to +5% for positive returns. Both figures show a symmetric structure between the negative and positive return exceedances. At the $\pm$0% threshold, both BTC/USD and BTC/EUR return and volume exceedances exhibit strong positive correlation, with the positive correlation for BTC/USD being slightly larger in magnitude. However, as we move to the extreme left and right tails, the correlation between return and volume exceedances declines and is only very weakly positive for both BTC/USD and BTC/EUR.

The estimation results for the univariate and bivariate models for ETH/USD and ETH/EUR return and volume exceedances are given in Tables 5 and 6 respectively. For Parts A and B in Tables 5 and 6, we notice that as we move towards the extreme left and right tails (through using larger fixed thresholds), the dependence parameter between the extreme returns and volumes increases, whilst the extreme correlation decreases. For the negative return exceedances of ETH/USD and ETH/EUR in Tables 5 and 6, the optimal

Table 2
Summary statistics of the de-trended traded volume of BTC/USD, BTC/EUR, ETH/USD, ETH/EUR for the period of 11:00 on 01/07/2017 to 11:00 on 20/04/2020, inclusive.

Statistics	BTC/USD	BTC/EUR	ETH/USD	ETH/EUR
Observations	24,434	24,434	24,434	24,434
Minimum	−6.056	−5.078	−6.556	−7.747
Q1	−0.514	−0.449	−0.616	−0.605
Median	−0.049	−0.030	−0.040	−0.036
Mean	0.000	0.000	0.000	0.000
Q3	0.467	0.418	0.578	0.578
Maximum	5.644	6.584	6.046	9.504
Skewness	0.360	0.280	0.231	0.219
Kurtosis	1.706	1.908	0.834	1.792
SD	0.812	0.740	0.962	0.980
Variance	0.659	0.548	0.926	0.960
Range	11.700	11.661	12.602	17.250
IQR	0.981	0.867	1.194	1.183

Table 3
Parameter estimation results for the univariate and bivariate models for BTC/USD return and volume exceedances. Thresholds in bold and labelled with * indicate the optimal threshold level computed by the procedure described in Section 2.1.1. These are given for the negative and positive returns on the last line in Parts A and B, respectively.

θ^{ret}	p^{ret}	σ^{ret}	ξ^{ret}	θ^{vol}	p^{vol}	σ^{vol}	ξ^{vol}	$\alpha^{ret/vol}$	$\rho^{ret/vol}$
Part A: Negative return exceedances and positive volume exceedances									
−5%	0.002	0.016	0.284	3.052	0.002	0.645	−0.022	0.999	0.002
		(0.004)	(0.183)			(0.157)	(0.194)	(0.000)	(0.000)
−4%	0.005	0.010	0.397	2.604	0.005	0.505	0.099	0.960	0.078
		(0.002)	(0.140)			(0.076)	(0.117)	(0.014)	(0.027)
−3%	0.010	0.010	0.215	2.201	0.010	0.469	0.095	0.922	0.149
		(0.001)	(0.069)			(0.045)	(0.072)	(0.014)	(0.026)
−2%	0.026	0.010	0.175	1.732	0.026	0.473	0.060	0.841	0.292
		(0.001)	(0.043)			(0.027)	(0.042)	(0.013)	(0.023)
−1%	0.076	0.009	0.144	1.168	0.076	0.533	−0.012	0.859	0.263
		(0.000)	(0.025)			(0.017)	(0.021)	(0.008)	(0.014)
−0%	0.497	0.006	0.000	−0.043	0.497	0.734	−0.106	0.417	0.826
		(0.000)	(0.003)			(0.008)	(0.006)	(0.000)	(0.000)
−1.97%*	0.027	0.009	0.177	1.718	0.027	0.478	0.054	0.935	0.126
		(0.001)	(0.043)			(0.027)	(0.041)	(0.008)	(0.015)
Part B: Positive return exceedances and positive volume exceedances									
+0%	0.503	0.006	0.000	−0.054	0.503	0.738	−0.108	0.382	0.854
		(0.000)	(0.003)			(0.008)	(0.006)	(0.000)	(0.000)
+1%	0.076	0.008	0.207	1.168	0.076	0.534	−0.012	0.909	0.174
		(0.000)	(0.027)			(0.017)	(0.021)	(0.007)	(0.013)
+2%	0.024	0.009	0.235	1.772	0.024	0.488	0.046	0.863	0.255
		(0.001)	(0.050)			(0.029)	(0.043)	(0.014)	(0.024)
+3%	0.009	0.012	0.214	2.261	0.009	0.480	0.091	0.895	0.198
		(0.001)	(0.085)			(0.049)	(0.077)	(0.019)	(0.034)
+4%	0.004	0.016	0.156	2.666	0.004	0.523	0.083	0.953	0.092
		(0.003)	(0.124)			(0.081)	(0.120)	(0.018)	(0.034)
+5%	0.002	0.020	0.082	3.002	0.002	0.593	0.031	0.950	0.097
		(0.004)	(0.151)			(0.131)	(0.177)	(0.024)	(0.047)
+1.85%*	0.028	0.009	0.210	1.710	0.028	0.476	0.055	0.932	0.132
		(0.001)	(0.045)			(0.027)	(0.041)	(0.009)	(0.017)

thresholds are −2.30% and −2.28%, respectively, for returns, and 1.994 and 2.005 for volume, with the extreme correlation being equal to 0.165 and 0.109, respectively. Overall, we see that the return value corresponding to the optimal negative threshold is smaller for ETH/USD, however, the extreme correlation is larger for ETH/USD. When considering the positive return exceedances of ETH/USD and ETH/EUR in Tables 5 and 6, the optimal thresholds are +2.20% and +2.19%, respectively, for returns, and 1.986 and 1.991 for volume, whilst the extreme correlation is equal to 0.099 and 0.103, respectively. We conclude that the return value corresponding to the optimal positive threshold is smaller for ETH/EUR, however, the extreme correlation is larger for ETH/EUR.

Figs. 3 and 4 provide a clearer illustration of the structure of the extreme correlation for fixed thresholds used to define extreme returns ranging from −5% to 0% for negative returns and 0% to +5% for positive returns. Both figures reveal a fairly symmetric structure between the negative and positive return exceedances. Similar to the results for the Bitcoin pairs, at the ±0% thresholds both ETH/USD and ETH/EUR return and volume exceedances exhibit strong positive correlation. However, as we move towards the extreme left and right tails, the correlation between return and volume exceedances decreases and becomes only very weakly positive. We observe at the extreme left tail, that the correlation between return and volume exceedances for both ETH/USD and ETH/EUR decreases at a similar rate. However, for the extreme right tail, the correlation between return and volume exceedances for ETH/EUR decreases much more quickly.

Overall, we conclude that all Bitcoin and Ethereum currency pairs exhibit similar results. Figs. 1 to 4 show that the correlation structure is quite symmetric considering both negative and positive return exceedances. Also in all cases, the correlation decreases when considering larger events in the left and right tails of the distributions. We observe at the ±0% threshold that trading in terms of the US Dollar compared with the Euro exhibits stronger positive correlation in terms of return and volume exceedances. In both cases, the optimal thresholds found for negative/positive tail exceedances, on average, lead to approximately 660–730 tail observations being used – reflecting a proportion of 2.7–3% of the total number of return observations (24,434). For all Bitcoin and Ethereum currency pairs, the tail indices corresponding to the optimal threshold values for both negative/positive return and volume exceedances are all positive. This suggests that the extreme return and volume exceedances can be described and modelled with a heavy tailed distribution.

5. Robustness checks

5.1. Daily returns and volumes

The results shown in Section 4 account only for high frequency (hourly) returns and volumes. To evaluate the robustness of our

Table 4
Parameter estimation results for the univariate and bivariate models for BTC/EUR return and volume exceedances. Thresholds in bold and labelled with * indicate the optimal threshold level computed by the procedure described in Section 2.1.1. These are given for the negative and positive returns on the last line in Parts A and B, respectively.

θ^{ret}	p^{ret}	σ^{ret}	ξ^{ret}	θ^{vol}	p^{vol}	σ^{vol}	ξ^{vol}	$\alpha^{ret/vol}$	$\rho^{ret/vol}$
Part A: Negative return exceedances and positive volume exceedances									
−5%	0.002	0.015	0.233	2.787	0.002	0.498	0.053	0.962	0.074
		(0.003)	(0.171)			(0.088)	(0.111)	(0.023)	(0.044)
−4%	0.005	0.013	0.209	2.406	0.005	0.483	0.054	0.946	0.105
		(0.002)	(0.108)			(0.062)	(0.089)	(0.018)	(0.034)
−3%	0.010	0.011	0.194	2.002	0.010	0.461	0.066	0.926	0.142
		(0.001)	(0.072)			(0.042)	(0.067)	(0.015)	(0.027)
−2%	0.026	0.010	0.174	1.552	0.026	0.460	0.048	0.926	0.143
		(0.001)	(0.044)			(0.026)	(0.041)	(0.009)	(0.016)
−1%	0.076	0.009	0.158	1.046	0.076	0.479	0.011	0.880	0.225
		(0.000)	(0.026)			(0.015)	(0.022)	(0.007)	(0.013)
−0%	0.494	0.006	0.000	−0.022	0.494	0.631	−0.080	0.500	0.750
		(0.000)	(0.003)			(0.006)	(0.005)	(0.000)	(0.000)
−1.90%*	0.030	0.010	0.183	1.503	0.030	0.452	0.054	0.940	0.117
		(0.001)	(0.043)			(0.025)	(0.039)	(0.007)	(0.013)
Part B: Positive return exceedances and positive volume exceedances									
+0%	0.506	0.005	0.000	−0.039	0.506	0.636	−0.081	0.502	0.748
		(0.000)	(0.003)			(0.006)	(0.005)	(0.000)	(0.000)
+1%	0.075	0.008	0.201	1.051	0.075	0.480	0.010	0.854	0.271
		(0.000)	(0.026)			(0.015)	(0.022)	(0.009)	(0.015)
+2%	0.023	0.009	0.278	1.607	0.023	0.472	0.038	0.947	0.104
		(0.001)	(0.053)			(0.028)	(0.042)	(0.008)	(0.016)
+3%	0.009	0.013	0.242	2.079	0.009	0.481	0.051	0.945	0.107
		(0.001)	(0.092)			(0.047)	(0.070)	(0.014)	(0.026)
+4%	0.004	0.021	0.000	2.447	0.004	0.521	0.023	0.973	0.053
		(0.003)	(0.099)			(0.068)	(0.083)	(0.014)	(0.027)
+5%	0.002	0.022	0.042	2.713	0.002	0.517	0.036	0.954	0.089
		(0.004)	(0.163)			(0.085)	(0.102)	(0.023)	(0.044)
+1.82%*	0.028	0.009	0.241	1.528	0.028	0.468	0.038	0.911	0.169
		(0.001)	(0.046)			(0.026)	(0.039)	(0.011)	(0.020)

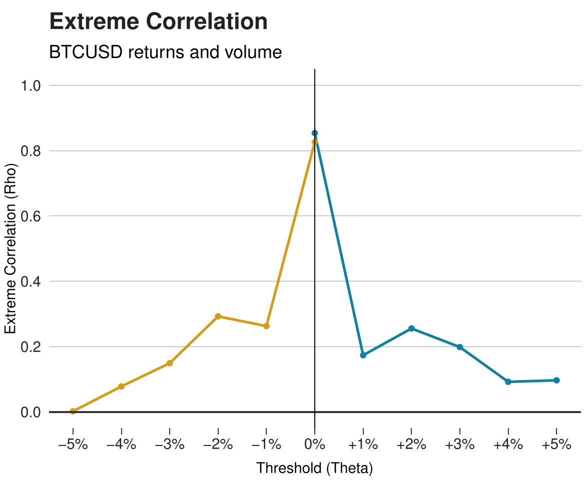

Fig. 1. Plots of the estimated extreme correlation $\rho^{ret/vol}$ between return and volume exceedances for BTC/USD for varying thresholds obtained from the estimation of the bivariate distribution (see results in Table 3).

results, we test whether the same features are seen in the daily returns and volumes of BTC/USD over the period of 2013–2020. As the market for Bitcoin has experienced extreme turbulent conditions in recent times, we can clearly identify three unique bull and bear market cycles that have occurred over the past few years. Therefore, to check the robustness of our analysis, we propose to check if the extreme correlation structure between return and volume is significantly different in these three bull and bear periods compared with the results found in Section 4. The three cycles correspond to the periods of: 6th October 2013 to 15th October 2015 (cycle 1), 16th October 2015 to 28th January 2019 (cycle 2), and 29th January 2019 to 21st July 2020 (cycle 3). To identify the three unique bull and bear cycles, we utilise the filtering method algorithm proposed by Lunde and Timmermann (2004). For further details regarding the algorithm, we refer the readers to Zhang et al. (2020) who provide a detailed example of this filtering method applied to high

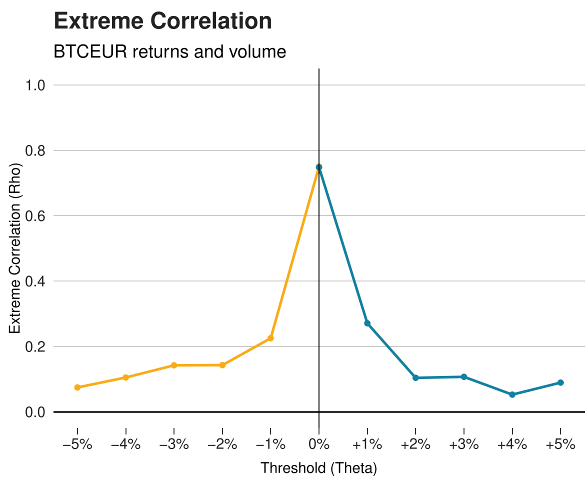

Fig. 2. Plots of the estimated extreme correlation $\rho^{ret/vol}$ between return and volume exceedances for BTC/EUR for varying thresholds obtained from the estimation of the bivariate distribution (see results in Table 4).

Table 5
Parameter estimation results for the univariate and bivariate models for ETH/USD return and volume exceedances. Thresholds in bold and labelled with * indicate the optimal threshold level computed by the procedure described in Section 2.1.1. These are given for the negative and positive returns on the last line in Parts A and B, respectively.

θ^{ret}	p^{ret}	σ^{ret}	ξ^{ret}	θ^{vol}	p^{vol}	σ^{vol}	ξ^{vol}	$\alpha^{ret/vol}$	$\rho^{ret/vol}$
Part A: Negative return exceedances and positive volume exceedances									
−5%	0.004	0.013	0.306	3.023	0.004	0.477	−0.022	0.971	0.058
		(0.002)	(0.127)			(0.065)	(0.086)	(0.015)	(0.030)
−4%	0.008	0.013	0.224	2.639	0.008	0.527	−0.053	0.967	0.065
		(0.001)	(0.081)			(0.049)	(0.058)	(0.012)	(0.023)
−3%	0.015	0.013	0.134	2.266	0.015	0.567	−0.068	0.977	0.045
		(0.001)	(0.051)			(0.038)	(0.041)	(0.007)	(0.013)
−2%	0.035	0.012	0.148	1.856	0.035	0.530	−0.028	0.933	0.130
		(0.001)	(0.037)			(0.025)	(0.033)	(0.008)	(0.016)
−1%	0.107	0.008	0.217	1.182	0.107	0.619	−0.073	0.874	0.236
		(0.000)	(0.024)			(0.016)	(0.016)	(0.007)	(0.012)
−0%	0.497	0.007	0.000	−0.034	0.497	0.886	−0.139	0.500	0.750
		(0.000)	(0.004)			(0.008)	(0.003)	(0.000)	(0.000)
−2.30%*	0.027	0.012	0.156	1.994	0.027	0.511	−0.011	0.914	0.165
		(0.001)	(0.043)			(0.029)	(0.040)	(0.011)	(0.020)
Part B: Positive return exceedances and positive volume exceedances									
+0%	0.503	0.007	0.000	−0.048	0.503	0.892	−0.140	0.435	0.810
		(0.000)	(0.004)			(0.008)	(0.003)	(0.000)	(0.000)
+1%	0.108	0.008	0.233	1.182	0.108	0.618	−0.072	0.927	0.141
		(0.000)	(0.023)			(0.016)	(0.016)	(0.006)	(0.011)
+2%	0.034	0.010	0.263	1.875	0.034	0.527	−0.025	0.921	0.153
		(0.001)	(0.044)			(0.026)	(0.034)	(0.010)	(0.018)
+3%	0.013	0.013	0.258	2.345	0.013	0.550	−0.060	0.948	0.102
		(0.001)	(0.072)			(0.039)	(0.045)	(0.011)	(0.021)
+4%	0.007	0.016	0.240	2.739	0.007	0.502	−0.037	0.961	0.076
		(0.002)	(0.112)			(0.052)	(0.066)	(0.014)	(0.026)
+5%	0.003	0.026	0.000	3.069	0.003	0.474	−0.020	0.977	0.046
		(0.005)	(0.138)			(0.067)	(0.090)	(0.015)	(0.029)
+2.20%*	0.028	0.010	0.265	1.986	0.028	0.010	0.265	0.949	0.099
		(0.001)	(0.048)			(0.001)	(0.048)	(0.008)	(0.015)

frequency cryptocurrency data.

Fig. 5 plots the extreme correlation $\rho^{ret/vol}$, between daily BTC/USD return and volume exceedances, for fixed thresholds used to define extreme returns ranging from −5% to 0% for negative returns and 0% to +5% for positive returns, for four samples (full sample period and the three cycle subperiods). It can be observed that all plots exhibit slightly different patterns. The plots of the full sample period and the cycle 2 subperiod show a fairly symmetric structure between the negative and positive return exceedances. In these two cases, at the ±0% threshold, return and volume exceedances exhibit strong positive correlation. However, as we move to the extreme left and right tails, the correlation between return and volume exceedances decreases and becomes only weakly positive. We notice that these results are similar to those seen in Section 4. In contrast, the plots of cycle 1 and cycle 3 in Fig. 5 reveal an asymmetric

Table 6
Parameter estimation results for the univariate and bivariate models for ETH/EUR return and volume exceedances. Thresholds in bold and labelled with * indicate the optimal threshold level computed by the procedure described in Section 2.1.1. These are given for the negative and positive returns on the last line in Parts A and B, respectively.

θ^{ret}	p^{ret}	σ^{ret}	ξ^{ret}	θ^{vol}	p^{vol}	σ^{vol}	ξ^{vol}	$\alpha^{ret/vol}$	$\rho^{ret/vol}$
Part A: Negative return exceedances and positive volume exceedances									
−5%	0.004	0.014	0.276	3.174	0.004	0.528	0.178	0.962	0.074
		(0.002)	(0.131)			(0.084)	(0.122)	(0.017)	(0.033)
−4%	0.008	0.015	0.163	2.749	0.008	0.557	0.100	0.959	0.081
		(0.002)	(0.074)			(0.057)	(0.072)	(0.013)	(0.025)
−3%	0.016	0.013	0.160	2.315	0.016	0.577	0.054	0.954	0.089
		(0.001)	(0.054)			(0.040)	(0.046)	(0.010)	(0.019)
−2%	0.035	0.012	0.156	1.849	0.035	0.568	0.045	0.928	0.140
		(0.001)	(0.038)			(0.027)	(0.033)	(0.008)	(0.016)
−1%	0.106	0.008	0.213	1.182	0.106	0.593	0.014	0.912	0.168
		(0.000)	(0.024)			(0.016)	(0.018)	(0.006)	(0.011)
−0%	0.496	0.007	0.000	−0.027	0.496	0.838	−0.079	0.497	0.753
		(0.000)	(0.004)			(0.008)	(0.003)	(0.000)	(0.000)
−2.28%*	0.027	0.013	0.132	2.005	0.027	0.570	0.050	0.944	0.109
		(0.001)	(0.041)			(0.031)	(0.037)	(0.009)	(0.016)
Part B: Positive return exceedances and positive volume exceedances									
+0%	0.504	0.007	0.000	−0.045	0.504	0.842	−0.080	0.500	0.750
		(0.000)	(0.003)			(0.008)	(0.003)	(0.000)	(0.000)
+1%	0.106	0.008	0.232	1.181	0.106	0.593	0.014	0.940	0.116
		(0.000)	(0.023)			(0.016)	(0.018)	(0.006)	(0.011)
+2%	0.033	0.010	0.295	1.882	0.033	0.566	0.048	0.989	0.022
		(0.001)	(0.045)			(0.028)	(0.034)	(0.004)	(0.007)
+3%	0.014	0.012	0.341	2.408	0.014	0.579	0.057	0.971	0.057
		(0.001)	(0.079)			(0.043)	(0.049)	(0.008)	(0.016)
+4%	0.006	0.019	0.187	2.874	0.006	0.557	0.108	0.958	0.083
		(0.003)	(0.099)			(0.068)	(0.083)	(0.014)	(0.028)
+5%	0.004	0.020	0.216	3.170	0.004	0.495	0.210	0.973	0.053
		(0.004)	(0.151)			(0.081)	(0.130)	(0.015)	(0.029)
+2.19%*	0.028	0.010	0.279	1.991	0.028	0.575	0.045	0.947	0.103
		(0.001)	(0.049)			(0.030)	(0.036)	(0.009)	(0.018)

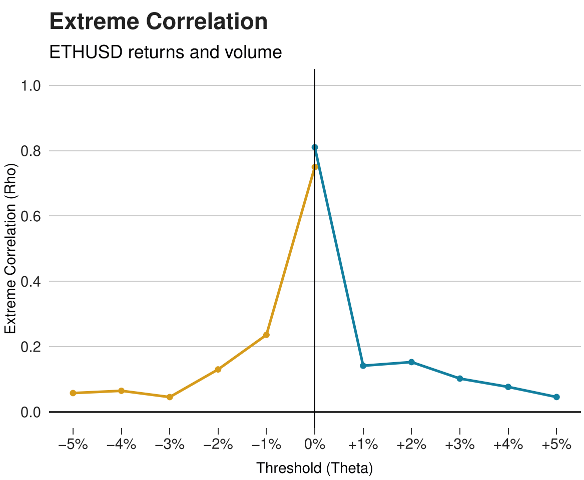

Fig. 3. Plots of the estimated extreme correlation $\rho^{ret/vol}$ between return and volume exceedances for ETH/USD for varying thresholds obtained from the estimation of the bivariate distribution (see results in Table 5).

structure between the negative and positive return exceedances. For cycle 1, we observe at the −0% threshold for negative return exceedances that return and volume exceedances show moderately strong correlation. However, as we move to the extreme left tail, the correlation between return and volume exceedances decreases and becomes less significant. Interestingly, the positive return exceedances for fixed thresholds ranging from 0% to +5% all exhibit no significant correlation with the corresponding volume exceedances. For cycle 3, we observe that negative return exceedances for fixed thresholds ranging from −0% to −5% all exhibit weak positive correlation with the corresponding volume exceedances. In contrast, positive return exceedances for fixed thresholds ranging from 0% to +5% all exhibit moderate correlation with the corresponding volume exceedances.

These results could be explained in relation to the publicity and popularity of Bitcoin during the three sample periods. The first

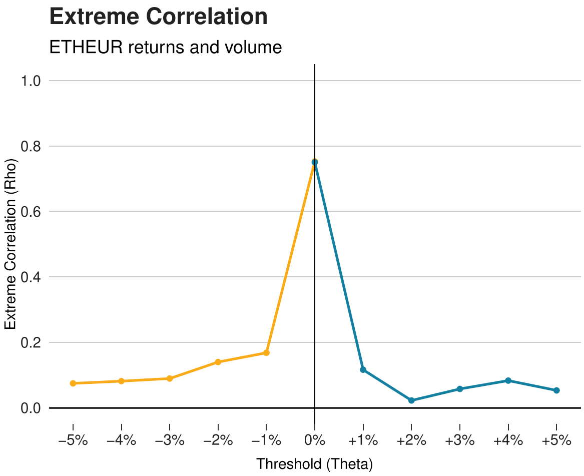

Fig. 4. Plots of the estimated extreme correlation $\rho^{ret/vol}$ between return and volume exceedances for ETH/EUR for varying thresholds obtained from the estimation of the bivariate distribution (see results in Table 6).

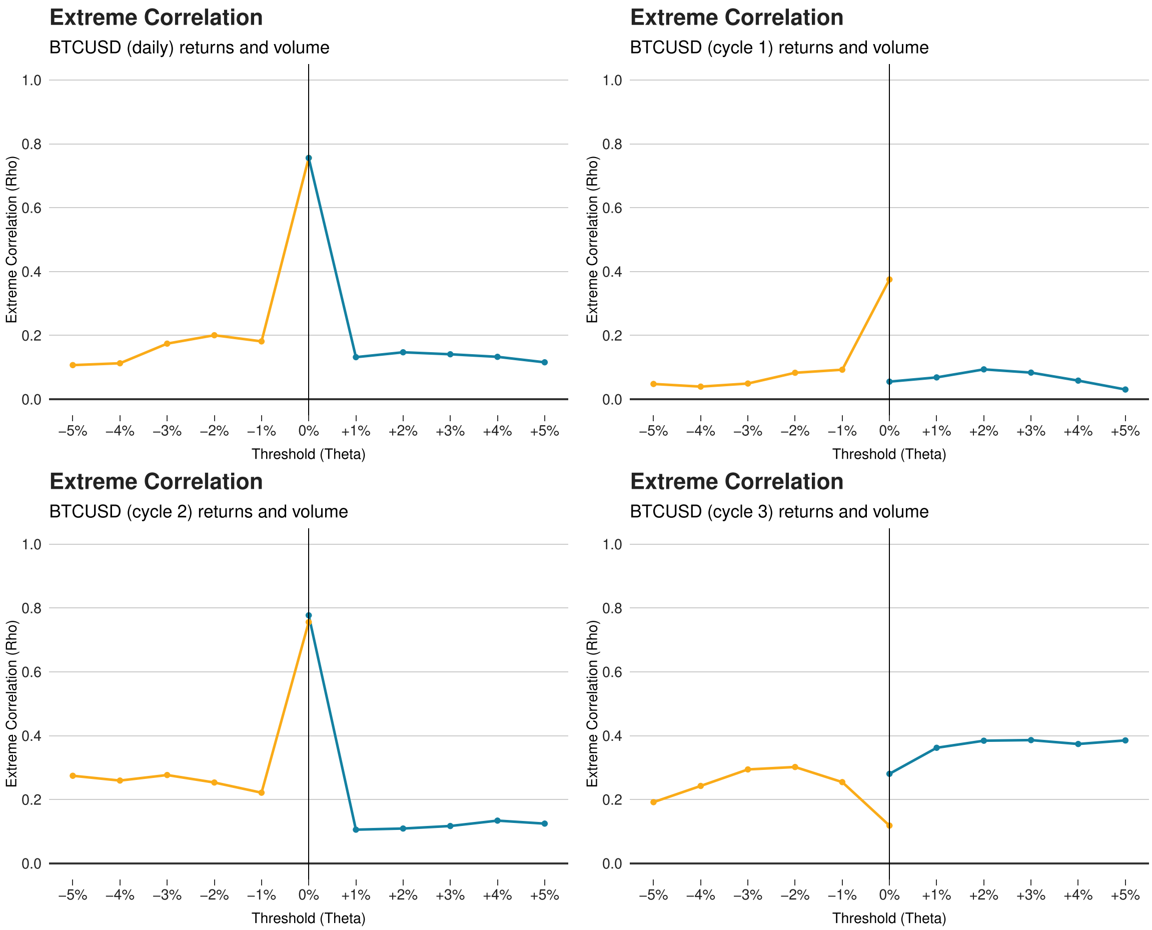

Fig. 5. Plots of the estimated extreme correlation $\rho^{ret/vol}$ between daily return and volume exceedances for varying thresholds for BTC/USD over the full sample period from 06/10/2013 to 21/07/2020 (top left); 06/10/2013 to 15/10/2015 (cycle 1) (top right); 16/10/2015 to 28/01/2019 (cycle 2) (bottom left); 29/01/2019 to 21/07/2020 (cycle 3) (bottom right).

period covers the early stages when interest in, and trading of, Bitcoin was still limited. During this period, the price of Bitcoin was relatively low and had only experienced one large spike at the start of the sample period before decreasing steadily until the end of the period. The magnitude of changes in the price, and thus returns, of Bitcoin were fairly small, and this was also the case for trading volume, which only started to increase towards the very end of the first period. However, significant price changes did not generally correspond with significant changes in volume, with volume remaining relatively stable even as the price changed over the period. The second period covers the explosion in the growth of Bitcoin and the rapid increase in awareness and interest between 2017 and 2018. During this period, Bitcoin saw significant increases and decreases in price, and also volume during these events. This similarity between the two variables reflected the increased interest, however, significant changes in trading volume were occurring frequently throughout the whole period. The third period covers a more recent and stable period for the price of Bitcoin. During this period, the price of Bitcoin also saw increases and decreases, however, these have been much more steady and less erratic compared with those in the second period. In addition, whilst significant changes in trading volume do seem to coincide with these large changes in the price, they also occur frequently during times when the price is fairly stable.

5.2. Geographical locations

For a further check of robustness, we analyse the impact of the geographical locations of cryptocurrency exchanges. The data obtained in Section 3 corresponds to the high frequency prices of Bitcoin and Ethereum listed on the Kraken cryptocurrency exchange (U.S. located exchange). We attempt to verify whether the results seen in Section 4 are similar to those for cryptocurrency exchanges located in other geographical regions. In this subsection, we apply the same methodology presented in Section 2 to the daily Bitcoin prices of: (i) BTC/USD from the Bitstamp exchange (Europe); (ii) BTC/USD from the OKCOIN exchange (Asia).

Fig. 6 plots the extreme correlation $\rho^{ret/vol}$ between daily return and volume exceedances of BTC/USD from both the Bitstamp and OKCOIN exchanges, for fixed thresholds used to define extreme returns ranging from −5% to 0% for negative returns and 0% to +5% for positive returns. We observe that both plots show similar results, where both the negative and positive return exceedances exhibit a fairly symmetric correlation structure. At the ±0% thresholds the return and volume exceedances from both exchanges show strong correlation, with the results from the European exchange showing a slighter stronger correlation than the Asian exchange. However, as we move to the extreme left and right tails, for both exchanges the correlation between return and volume exceedances decreases towards moderate positive correlation. These results are found to be similar to those seen in Fig. 4.

Overall, from the results of our robustness checks it is clear that the structure of the extreme correlation of returns and volumes of Bitcoin (in terms of the US Dollar) at a daily frequency and from exchanges located in other geographical areas are similar to those observed in Section 4 for high frequency intervals. Hence, we believe that our results are generally robust to the different specifications of data frequency and geographical location of exchanges.

6. Discussion

As the cryptocurrency market is still in the early stages of its development, one may hypothesise that it exhibits similar properties to emerging markets (in banking or technology) or those from the pre-market-efficiency era. Our empirical study provides an essential stepping stone to further understand the economic and financial perspective in acknowledging how market information is conveyed and incorporated in cryptocurrency prices.

As stated in Section 1, there currently exists little or no literature on the extreme dependence between the volumes and returns of cryptocurrencies in a high frequency setting. The fact that investors and traders rely heavily on models based on trading rules and the relationship between return and volume further highlights the importance for a better understanding of the volume–return relationship in cryptocurrencies.

In general, the literature on the stylised facts of volume–return suggests that there is positive correlation between return and the

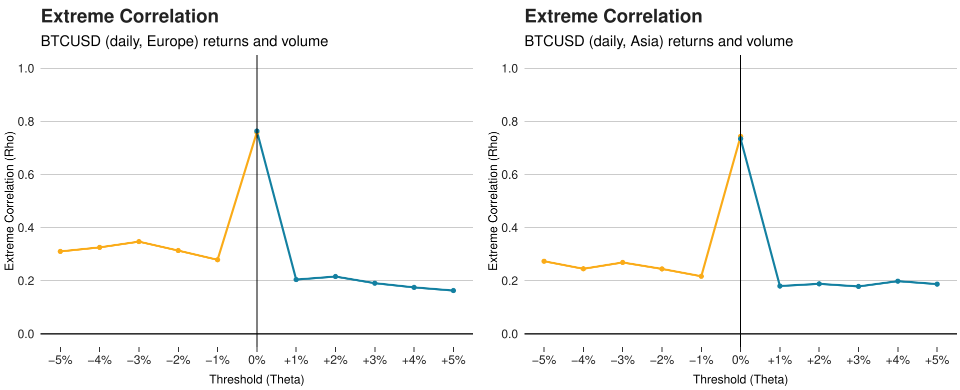

Fig. 6. Plots of the estimated extreme correlation $\rho^{ret/vol}$ between daily return and volume exceedances for varying thresholds for BTC/USD (Europe) (left) and BTC/USD (Asia) (right).

absolute return volume (Karpoff, 1987) in equities (Ying, 1966; Jain and Joh, 1988; Llorente et al., 2002), interest rates and currency futures (Puri and Philippatos, 2008) and real estate (Tsai, 2014). Hence, these markets appear to go against the efficient market hypothesis. In these cases, market participants hold private information about the liquidation value of these assets, hence they will position their demand for the asset proportionally to the signal received. Therefore, when they receive news about an extreme change in the liquidation value of the asset, they will trade more aggressively (De Bondt and Thaler, 1985).

However, our paper appears to contradict these results as we observe that for Bitcoin and Ethereum there exists weak positive correlation between return and volume when considering extreme events in the left and right tails of the distributions. Our results are generally robust to different specifications of data frequency and geographical location of exchanges. From an economic perspective, this result suggests that misinterpretation among market participants can lead to cryptocurrency markets becoming relatively illiquid, resulting in extreme price movements. Similar results are also observed in financial indices as shown by Gennotte and Leland (1990), and Longin and Pagliardi (2016).

7. Conclusion

In this paper, we investigate the extreme dependence between high frequency cryptocurrency returns and volume at the extreme tails, and analyse the extreme correlation between the two variables. Our analysis utilised the returns and volume of Bitcoin and Ethereum in terms of the US Dollar and the Euro, over a three year period from July 2017 to April 2020, inclusive. To the best of our knowledge, this is the first time that the high frequency extreme dependence of cryptocurrency volume and returns at the extreme tails has been investigated.

The analysis of the extreme dependence was conducted using extreme value theory (peaks over threshold method), and highlights how these results assist in trading strategies for traders and practitioners during extreme events. The results indicate that the correlation structures of both negative and positive extreme return and volume exceedances in Bitcoin and Ethereum are similar and symmetric. At the $\pm 0\%$ thresholds, both Bitcoin and Ethereum currency pairs exhibit strong positive correlation in terms of return and volume exceedances. However, in all cases the correlation decreases and becomes weakly positive when considering extreme events in the left and right tails of the distributions. Our robustness checks reveal similar outcomes when considering the structure of the extreme correlation of Bitcoin (in terms of the US Dollar) returns and volumes at a daily frequency and for cryptocurrency exchanges located in other geographical areas such as Europe and Asia. These results are in contrast to those obtained in the survey by Karpoff (1987), which suggest significant positive correlation between return and the absolute return volume at the extremes.

Our results from an extreme value perspective provide a platform to further understand the economic and financial perspective in acknowledging how market information is conveyed and incorporated in cryptocurrency prices. From an economic perspective, our results suggest there exists weak positive correlation between return and volume in the tails, which implies that a misinterpretation among market participants can cause cryptocurrencies markets to become relatively illiquid, leading to extreme price movements. Relating our statistical findings to economic models, we find that our empirical results are consistent with the explanation of market crashes by Gennotte and Leland (1990) and Longin and Pagliardi (2016) based on trade misinterpretation.

Availability of data and material (data transparency)

Not applicable.

Code availability

Not applicable.

Funding

Not applicable.

Authors' contribution

Chan Stephen, Chu Jeffrey and Zhang Yuanyuan: conceptualisation, methodology, software, validation, formal analysis, investigation, data curation, writing – original draft, writing – review & editing, visualisation, project administration. Nadarajah Saralees: software.

Conflict of interest

None declared.

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